

GPU Node

Troubleshooting & Disk Management

Technical Documentation — Session: January 23, 2026

Cluster: aidc-artgarage HPC | Nodes: aidc-artgarage · hpcm-vm · gpu01

The Problem: When `cm image capture --node gpu01 -i rhel8.10-gpu01-23012026` was run on hpcm-vm, it uses `rsync` to copy the entire gpu01 filesystem to hpcm-vm. But gpu01 had two massive directories:

- `/var/lib/docker/overlay2` — **456 GB** of Docker image layers
- `/var/spool/pbs/mom_logs` — **63 GB** of PBS scheduler logs (6 months worth)
- hpcm-vm's root disk `/dev/vda31` is only **117 GB**. So `rsync` tried to pour ~519 GB into a 117 GB disk and failed with:

`rsync: write failed: No space left on device (28)`

1. Overview and Cluster Environment

This document captures the full troubleshooting session of 23 January 2026 on the aidc-artgarage HPC cluster at IISc ArtGarage, Bengaluru.

Every command shown below is taken verbatim from those logs.

The session involved: disk-space investigation on gpu01, Docker image cleanup, PBS job monitoring; real-time disk monitoring on hpcm-vm; and HPCM cm image capture / delete / verify on hpcm-vm.

Component	Detail
Admin node (login)	aidc-artgarage — ext: 172.25.60.80 mgmt: 10.1.1.101 RHEL 8.10
HPCM management VM	hpcm-vm — IP: 10.1.1.100 /dev/vda31: 117 GB root FS
GPU compute node	gpu01 — IP: 10.1.1.110 Software RAID /dev/md0 2 TB+ storage
IB fabric	Mellanox ConnectX (mlx5_core / mlx4) — o2ib InfiniBand RDMA
Lustre FS	Servers: 10.128.1.202-205@o2ib → /lustre1 (/home mount)
Job scheduler	PBS Pro 2025.1.0 (pbs_server @ aidc-artgarage:15001)
Container runtime	Docker 24.x — overlay2 storage driver
Log source tool	MobaXterm Personal Edition v25.4

2. Terminal Session – File 1: aidc-artgarage & gpu01

Source: MobaXterm_root_gpu01varspoolpbsmom_logs_20260123_152527.rtf

2.1 Initial System Assessment on aidc-artgarage

```
[root@aidc-artgarage ~]# top
top - 11:42:16 up 2 days, 19:12, 3 users, load average: 0.71, 2.54, 4.78
Tasks: 2873 total, 1 running, 2872 sleeping, 0 stopped, 0 zombie
%Cpu(s): 0.7 us, 0.3 sy, 0.0 ni, 99.0 id, 0.0 wa, 0.1 hi, 0.0 si, 0.0 st
MiB Mem : 515220.3 total, 96969.3 free, 249381.9 used, 168869.1 buff/cache
MiB Swap: 24576.0 total, 24571.7 free, 4.2 used. 260131.5 avail Mem

  PID USER      PR  NI   VIRT   RES   SHR S  %CPU  %MEM    TIME+    COMMAND
 52110 qemu     20   0 65.9g  59.7g 23660 S  16.2  11.9 716:16.02 qemu-kvm
 3803 root       20   0 6100240 573108 141004 S   2.0   0.1 101:47.06 k3s-server
2477613 root     20   0 265532   7608  3568 R   1.0   0.0  0:06.20 top
```

[i] Load is 0.71 – system is idle. qemu-kvm (hpcm-vm) uses 59.7 GB RAM. k3s-server active. No IB or Lustre problems visible.

```
[root@aidc-artgarage ~]# ls
20251222  bm_package_v3.zip  gol.24.4.linux-amd64.tar.gz  cluster-configfile
gpu01-properties  altair_lic.dat  cmd-27-07-2025.txt  gpu-test-script
Documents  ice.conf  Downloads  initial-setup-ks.cfg  backup_20250910  etc  intel
```

```
[root@aidc-artgarage ~]# ll /home/soft/gpu-node-backup/
total 4
drwxr-xr-x 21 root root 4096 Jan 23 12:19 rhel8.10-gpu01-27072025
```

```
[root@aidc-artgarage ~]# du -sh /home/soft/gpu-node-backup/rhel8.10-gpu01-27072025
130G    /home/soft/gpu-node-backup/rhel8.10-gpu01-27072025
```

[i] An old 130 GB backup of gpu01 (rhel8.10-gpu01-27072025) exists on login node under /home/soft/gpu-node-backup/.

2.2 SSH to gpu01 – Disk Investigation

```
[root@aidc-artgarage ~]# ssh gpu01
Last login: Fri Jan 23 11:45:19 2026 from 10.1.1.110
```

```
[root@gpu01 ~]# ls
20251222  install-compute.sh  pbs-gpu01.zip  pbs_mom_20250728  pbspro-execution-
```

```
[root@gpu01 ~]# df -h
```

Filesystem	Size	Used	Avail	Use%	Mounted on
devtmpfs	1009G	0	1009G	0%	/dev
tmpfs	1009G	0	1009G	0%	/dev/shm
/dev/md0p31	1.8T	1.2T	575G	68%	/
tmpfs	202G	3.5G	199G	2%	/run
/dev/md0p11	1014M	248M	633M	29%	/boot
tmpfs	202G	82M	202G	1%	/run/user/0
10.128.1.202@o2ib,10.128.1.203@o2ib:10.128.1.204@o2ib,10.128.1.205@o2ib:/lustre1					(mounted)

[i] IB confirmed: Lustre is mounted via @o2ib RDMA. /dev/md0p31 root at 68%. All four IB storage servers reachable.

```
[root@gpu01 ~]# cd /
(changed directory to /)
```

```
[root@gpu01 /]# ls
bin boot dev etc home lib lib64 media mnt opt proc root run sbin srv sys tmp usr
var
```

```
[root@gpu01 /]# du - sh media
du: cannot access '-': No such file or directory
du: cannot access 'sh': No such file or directory
0      media
```

✗ Typo: space between '-' and 'sh' splits them into two separate arguments. Correct syntax: `du -sh <path>`

```
[root@gpu01 /]# du -sh media
0      media
```

```
[root@gpu01 /]# du -sh mnt
0      mnt
```

```
[root@gpu01 /]# du -sh opt
1.1G   opt
```

```
[root@gpu01 /]# du -sh var
519G   var
```

```
[root@gpu01 /]# cd /var/
(changed to /var/)
```

```
[root@gpu01 var]# du -sh *
0      adm
117M   cache
4.0K   clustertest
0      crash
0      db
0      empty
0      ftp
0      games
0      gopher
0      kerberos
456G   lib      <— Docker images live here
0      local
0      lock
35M    log
0      mail
16K    named
0      nis
0      opt
0      preserve
0      run
63G    spool    <— PBS mom_logs live here
44K    tmp
16K    yp
```

✗ CRITICAL: `/var/lib` = 456 GB (Docker overlay2) and `/var/spool` = 63 GB (PBS mom_logs). Both will overflow hpcm-vm's 117 GB disk during cm image capture.

```
[root@gpu01 ~]# du -sh /var/lib/docker
456G   /var/lib/docker
```

2.3 PBS Job Monitoring – aidc-artgarage

```
[root@aidc-artgarage ~]# qstat -ans
```

aidc-artgarage:

Job ID	Username	Queue	Jobname	SessID	NDS	TSK	Req'd Memory	Req'd Time	Elap S	Time
5277.aidc-artg*	prakhar	workq	pi0.5-fin*	27385	1	24	144gb	1500:	R	01:34
gpu01/0*24 Job run at Fri Jan 23 at 11:19 on (gpu01:ncpus=24:ngpus=2:mem=150994944kb)										
5279.aidc-artg*	sujith	workq	fiall	28961	1	16	128gb	25600	R	01:22
gpu01/1*16 Job run at Fri Jan 23 at 11:31 on (gpu01:ncpus=16:ngpus=4:mem=134217728kb)										

[i] Two active jobs on gpu01: Job 5277 (pi0.5-finetune) — 24 CPUs, 2 GPUs, 144 GB. Job 5279 (fiall) — 16 CPUs, 4 GPUs, 128 GB. Both must not be disturbed.

```
[root@aidc-artgarage ~]# qstat -xf 5277.aidc-artg*
```

qstat: illegally formed job identifier: 5277.aidc-artg*

✗ Wildcards are not valid in qstat job identifiers. Use the full numeric job ID instead.

```
[root@aidc-artgarage ~]# qstat -xf 5277
```

```
Job Id: 5277.aidc-artgarage
Job_Name = pi0.5-finetune
Job_Owner = prakhar@aidc-artgarage.hpc.com
resources_used.cput = 0
resources_used.cput = 00:00:00
resources_used.mem = 12360kb
resources_used.ncpus = 24
resources_used.vmem = 151968kb
resources_used.walltime = 01:36:36
job_state = R
queue = workq
server = aidc-artgarage
exec_host = gpu01/0*24
exec_vnode = (gpu01:ncpus=24:ngpus=2:mem=150994944kb)
interactive = True
Resource_List.mem = 144gb
Resource_List.ncpus = 24
Resource_List.ngpus = 2
Submit_arguments = -I -l select=1:ncpus=24:ngpus=2:mem=144GB -l walltime=1500:00:00 -N pi0.5-finetune
```

```
[root@aidc-artgarage ~]# qstat -ans
```

(same output as above – both jobs still R state at elapsed time 01:38 / 01:26)

2.4 Docker Cleanup on gpu01 – Round 1

```
[root@gpu01 ~]# docker ps
```

CONTAINER ID	IMAGE	COMMAND	CREATED	STATUS	PORTS	NAMES
(no running containers)						

```
[root@gpu01 ~]# docker images
```

REPOSITORY	TAG	IMAGE ID
aidc-artgarage:5000/perception_super_app_docker	latest	a72dc4310fa8
aidc-artgarage:5000/perception_super_app_docker	<none>	a6f82a11e2eb
aidc-artgarage:5000/perception_super_app_docker	<none>	da9bec4135d8
aidc-artgarage:5000/multimatter-openpi	latest	c935d8a60a1e
aidc-artgarage:5000/multimatter-openpi	<none>	e4fc728c264a
aidc-artgarage:5000/multimatter-openpi	<none>	5565b1a61e01
aidc-artgarage:5000/multimatter-openpi	<none>	3d59fcbbcd1f
aidc-artgarage:5000/multimatter-openpi	<none>	c2b4fa2f93b7
aidc-artgarage:5000/multimatter-openpi	<none>	da6702bd863c

... (26 images total)

```
[root@gpu01 ~]# docker rmi a6f82a11e2eb
```

```
Untagged: aidc-artgarage:5000/perception_super_app_docker@sha256:a019e257f33dc33...
Deleted: sha256:a6f82a11e2ebd4ff5ac9976d8b1405e23b398c910e8412e68692a25df9653fde
Deleted: sha256:a2f1638a1fbcb29382df9ca21f9af38c35fe1ad01feaea839fb373ee7022729e
```

```
[root@gpu01 ~]# df -h
```

(df after each rmi – Lustre mount via o2ib still visible and healthy)

```
[root@gpu01 ~]# docker rmi e4fc728c264a 5565b1a61e01 3d59fcbbcd1f c2b4fa2f93b7
```

```
Untagged: aidc-artgarage:5000/multimatter-openpi@sha256:d03054eeb261737577b0cf20...
Deleted: sha256:e4fc728c264ada6286616136874339eb16d1ce5b60ab388b1c3807bc50a77e0c
Untagged: aidc-artgarage:5000/multimatter-openpi@sha256:ce83786797fe8aec73f3cf47...
Deleted: sha256:5565b1a61e01698640d1607e699331cdc09bfae78d6786b0c53b04253afb372d
Deleted: sha256:3d59fcbbcd1f0ba674475728521afa8b3a7f5978afb50e3caldae6fc93d6267
Deleted: sha256:c2b4fa2f93b7...
```

```
[root@gpu01 ~]# docker rmi cf58f698fe91 80be2058e2ab 53d4f782ecd5 565968b73a61
```

```
Untagged: robot-lab:latest
Deleted: sha256:53d4f782ecd5ac88685902c6792ab70428ddbdcdb9eac3be9cb1d886a1d23
Deleted: sha256:30df07f61093bc49130f30ecb2a5fd92f934c899923b9786af81a254ee3a4e7d
Deleted: sha256:b2af7a6c149a25044ab83046e52454be1f49a291d72c0cb8b4404704fea79fe3
Deleted: sha256:37b6a5952435e4c90b9a438716b69bb70fc57a118a54963029f5bca00666c828
```

```
[root@gpu01 ~]# docker rmi e680c1c45ecd
```

```
Untagged: aidc-artgarage:5000/pytorch:22.02-py3
Untagged: aidc-artgarage:5000/pytorch@sha256:8b8fe38fd76065aa8c67b9395975aa2102c...
Deleted: sha256:e680c1c45ecd32f7e7a0ee117bfe8b6206e68e4ab3cdac1ceb5c2676d666a9a
```

```
[root@gpu01 ~]# docker rmi 5a812786d99b 2740565fb3f0 224c963be240 ef4c5017539c
```

```
Untagged: aidc-artgarage:5000/chetan/test_gpu:latest
Untagged: aidc-artgarage:5000/chetan/test_gpu@sha256:28b685c8abca92cad5efec14843...
Deleted: sha256:5a812786d99bc76b89d866ba76b8dade3c5927227943f3e4f1c88b4e735f46cc
Deleted: sha256:2b97831893c85ff455d4d6e1326564fedd05ec60fcdc15eae3ee36324e4febb8
```

2.5 Multi-Tag Docker Conflict and Resolution

```
[root@gpu01 ~]# docker rmi c73c55d3f8b9 c73c55d3f8b9
```

```
Error response from daemon: conflict: unable to delete c73c55d3f8b9 (must be forced)
Error response from daemon: conflict: unable to delete c73c55d3f8b9 (must be forced)
```

✗ Same image ID tagged under two different repository names. Must remove each named tag separately.

```
[root@gpu01 ~]# docker images --no-trunc | grep c73c55d3f8b9
```

```
aidc-artgarage:5000/tensorflow      25.02-tf2-py3  sha256:c73c55d3f8b...
aidc-artgarage:5000/chetan/tensorflow 25.02-tf2-py3  sha256:c73c55d3f8b...
```

[i] Two named tags share the same image ID. Resolution: docker rmi each tag individually.

```
[root@gpu01 ~]# docker rmi aidc-artgarage:5000/tensorflow:25.02-tf2-py3
```

```
Untagged: aidc-artgarage:5000/tensorflow:25.02-tf2-py3
Untagged: aidc-artgarage:5000/tensorflow@sha256:3f90bb9ec86f6384c2cafdeeda06aafa...
```

```
[root@gpu01 ~]# docker rmi aidc-artgarage:5000/chetan/tensorflow:25.02-tf2-py3
```

```
Error response from daemon: conflict: unable to remove repository reference – container using image
```

✗ A stopped container still holds a reference to this image. Must remove all stopped containers first.

2.6 Stop and Remove All Stopped Containers

```
[root@gpu01 ~]# docker ps a
```

```
"docker ps" accepts no arguments. See 'docker ps --help'.
Usage:  docker ps [OPTIONS]
List containers
```

✗ Typo: 'a' is not a valid argument to docker ps. Correct flag is: docker ps -a

```
[root@gpu01 ~]# docker ps aa
```

```
"docker ps" accepts no arguments. See 'docker ps --help'.
```

```
[root@gpu01 ~]# docker ps -a
```

CONTAINER ID	IMAGE	COMMAND
e224221ea6c3	27dede7dfab2	"/opt/nvidia/nvidi..."
eeaa458bd0cc	isaac-lab-strider	"bash"
e33491ab4e59	aidc-artgarage:5000/cleargrasp-train:latest	"/opt/nvidia/nvidi..."
af693c12bcea	aidc-artgarage:5000/cleargrasp-train:latest	"/opt/nvidia/nvidi..."
e8a7e01cdd11	aidc-artgarage:5000/cleargrasp-train:latest	"/opt/nvidia/nvidi..."
06a58b9143a9	aidc-artgarage:5000/tensorflow:25.02-tf2-py3	"/opt/nvidia/nvidi..."
5fd4b8fccb71	aidc-artgarage:5000/ultralytics/ultralytics	"/bin/bash"
6b5024a4ae91	aidc-artgarage:5000/ultralytics/ultralytics	"/bin/bash"
0f7db9d3112a	nvcr.io/nvidia/isaac-sim:4.5.0	"/bin/sh -c /isaa..."
6595c5409775	...	
7a97bc73c6de	...	
6db5fc269d7a	...	
5bf64f3c0864	...	
4d09dal1f674e	...	

(14 stopped containers total)

```
[root@gpu01 ~]# docker stop e224221ea6c3 eeaa458bd0cc af693c12bcea e33491ab4e59
```

```
e224221ea6c3
eeaa458bd0cc
af693c12bcea
e33491ab4e59
e8a7e01cdd11
06a58b9143a9
6b5024a4ae91
5fd4b8fccb71
```

```
[root@gpu01 ~]# docker stop 6b5024a4ae91 0f7db9d3112a 6595c5409775 7a97bc73c6de
```

```
6b5024a4ae91
0f7db9d3112a
6595c5409775
7a97bc73c6de
6db5fc269d7a
5bf64f3c0864
4d09dal1f674e
```

```
[root@gpu01 ~]# docker rm $(docker ps -aq)
```

```
e224221ea6c3
eeaa458bd0cc
e33491ab4e59
af693c12bcea
e8a7e01cdd11
06a58b9143a9
5fd4b8fccb71
6b5024a4ae91
0f7db9d3112a
6595c5409775
7a97bc73c6de
6db5fc269d7a
5bf64f3c0864
4d09dal1f674e
```

✓ All 14 stopped containers removed. 867.1 MB freed; blocked image deletions now unblocked.

2.7 Continue Docker Image Removal – Round 2

```
[root@gpu01 ~]# w
13:15:13 up 4:52, 1 user, load average: 4.43, 4.62, 4.54
USER      TTY      FROM          LOGIN@      IDLE JCPU PCPU WHAT
root      pts/1    10.1.1.101    13:14      0.00s 0.01s 0.01s w
```

```
[root@gpu01 ~]# docker images
REPOSITORY                                     TAG          IMAGE ID
aidc-artgarage:5000/perception_super_app_docker latest       a72dc4310fa8
aidc-artgarage:5000/perception_super_app_docker <none>       da9bec4135d8
aidc-artgarage:5000/multimatter-openpi        latest       c935d8a60a1e
aidc-artgarage:5000/multimatter-lerobot        latest       8937ce116f9a
aidc-artgarage:5000/multimatter-lerobot        <none>       5cfc843944d4
aidc-artgarage:5000/multimatter-lerobot        <none>       0da0f70a5d5e
aidc-artgarage:5000/multimatter-lerobot        <none>       7c35907cd167
aidc-artgarage:5000/multimatter-lerobot        <none>       f49a0432af56
aidc-artgarage:5000/multimatter-lerobot        <none>       929ec38cc257
... (26 images remaining)
```

```
[root@gpu01 ~]# docker system df
TYPE                TOTAL    ACTIVE    SIZE      RECLAIMABLE
Images              26       6        271.6GB   160GB (58%)
Containers          14       0         867.1MB   867.1MB (100%)
Local Volumes       0        0          0B        0B
Build Cache        123      0        1.773GB   1.773GB
```

i docker system df: 160 GB reclaimable from images (58%), 867.1 MB from containers (100%), 1.773 GB build cache.

```
[root@gpu01 ~]# docker rmi c73c55d3f8b9 c73c55d3f8b9
Error response from daemon: conflict: unable to delete c73c55d3f8b9 (must be forced)
Error response from daemon: conflict: unable to delete c73c55d3f8b9 (must be forced)
```

```
[root@gpu01 ~]# docker ps
CONTAINER ID   IMAGE                                COMMAND                  CREATED        STATUS        PORTS        NAMES
```

```
[root@gpu01 ~]# docker rmi aidc-artgarage:5000/tensorflow:25.02-tf2-py3
Untagged: aidc-artgarage:5000/tensorflow:25.02-tf2-py3
Untagged: aidc-artgarage:5000/tensorflow@sha256:3f90bb9ec86f6384c2cafdeeda06aafa...
```

```
[root@gpu01 ~]# docker rmi aidc-artgarage:5000/chetan/tensorflow:25.02-tf2-py3
Error response from daemon: conflict: unable to remove repository reference "...f8b9"
```

```
[root@gpu01 ~]# qstat -ans
(Jobs 5277 and 5279 still running at elapsed 02:06 and 01:54 respectively)
```

```
[root@gpu01 ~]# docker rmi a72dc4310fa8 da9bec4135d8
Untagged: aidc-artgarage:5000/perception_super_app_docker:latest
Deleted: sha256:a72dc4310fa8addbb3ab7e6f40a073b9aa84ad17f70bd0105ddeb97aefa6dae9
Deleted: sha256:ec963c381b478e1d438e1eb2f6b0853dcef913341a1ce8639dc5faf79001da6c
Deleted: sha256:2068b36c3d82c25901a2c5c9e73d9f8c05c2e9101676d708630cd39085bc7111
```

```
[root@gpu01 ~]# docker images
(perception_super_app_docker rows gone; multimatter-openpi + lerobot remain)
```

```
[root@gpu01 ~]# docker rmi c935d8a60a1e
Untagged: aidc-artgarage:5000/multimatter-openpi:latest
Deleted: sha256:c935d8a60a1e2ea525b389d9b2c58a903d7630957d20c0ce2f0c8158263c2464
Deleted: sha256:f72bcce0f7c0683f067fdbc20e0ec7bb2588d8d83458c51fde1e539e5d4dc0c6
```



```
[root@gpu01 ~]# docker rmi 8937ce116f9a 5cfc843944d4 0da0f70a5d5e 7c35907cd167
```

```
Untagged: aidc-artgarage:5000/multimatter-lerobot:latest
Deleted: sha256:8937ce116f9a5f5199c4522a2b773ffb4f254ee0824311aaf24474bca95aa2b3
Deleted: sha256:4308710008bc1301be60cfe612211b508f6439c7369919e85d7c9d5444b50af6
Deleted: sha256:5cfc843944d4bda2bb6a058a6968d9032960cc0ac515af1a9e676554f8501264
Deleted: sha256:0da0f70a5d5e...
Deleted: sha256:7c35907cd167...
```

```
[root@gpu01 ~]# docker rmi c73c55d3f8b9 c73c55d3f8b9
```

```
Error response from daemon: conflict: unable to delete c73c55d3f8b9 (must be forced)
Error response from daemon: conflict: unable to delete c73c55d3f8b9 (must be forced)
```

```
[root@gpu01 ~]# docker images
```

aidc-artgarage:5000/multimatter-lerobot	<none>	8791df39c63e	4 weeks ago
aidc-artgarage:5000/multimatter-lerobot	<none>	6ec7a9230a07	4 weeks ago
aidc-artgarage:5000/multimatter-lerobot	<none>	6645f51ac6a2	5 weeks ago
aidc-artgarage:5000/multimatter-lerobot	<none>	1b3e5202a4a0	5 weeks ago
aidc-artgarage:5000/multimatter-lerobot	<none>	9799c129e6ee	5 weeks ago
aidc-artgarage:5000/multimatter-lerobot	<none>	27dede7dfab2	5 weeks ago
aidc-artgarage:5000/cleargrasp-train	latest	80be2058e2ab	3 months ago
isaac-lab-strider	latest	cf58f698fe91	4 months ago
aidc-artgarage:5000/ultralytics/ultralytics	latest	75f29cdb37cc	5 months ago
aidc-artgarage:5000/chetan/tensorflow	25.02-tf2-py3	c73c55d3f8b9	11 months ago
nvr.io/nvidia/isaac-sim	4.5.0	ca34eb4ab11f	12 months ago

```
[root@gpu01 ~]# docker rmi 8791df39c63e 6ec7a9230a07 6645f51ac6a2 1b3e5202a4a0
```

```
Untagged/Deleted: sha256:8791df39c63ee6366cad4c0d7ffcf975215ebdb1b47de2c3aa9cf42d27cf4d15
Deleted: sha256:d53f9ea9869e31c5e9d1389cfc1e16dbf3e0cdad7f31720dba5078218106476f
Deleted: sha256:6ec7a9230a07...
Deleted: sha256:6645f51ac6a2...
Deleted: sha256:1b3e5202a4a0...
```

```
[root@gpu01 ~]# docker ps -a
```

CONTAINER ID	IMAGE	COMMAND
e224221ea6c3	27dede7dfab2	"/opt/nvidia/nvidia..."
eeaa458bd0cc	isaac-lab-strider	"bash"
e33491ab4e59	aidc-artgarage:5000/cleargrasp-train:latest	"/opt/nvidia/nvidia..."
af693c12bcea	aidc-artgarage:5000/cleargrasp-train:latest	"/opt/nvidia/nvidia..."
e8a7e01cdd11	aidc-artgarage:5000/cleargrasp-train:latest	"/opt/nvidia/nvidia..."
06a58b9143a9	c73c55d3f8b9 (tensorflow)	"/opt/nvidia/nvidia..."
5fd4b8fccb71	aidc-artgarage:5000/ultralytics/ultralytics	"/bin/bash"
0f7db9d3112a	nvr.io/nvidia/isaac-sim:4.5.0	"/bin/sh -c /isaa..."
... (14 stopped containers)		

```
[root@gpu01 ~]# docker rm $(docker ps -aq)
```

```
e224221ea6c3
eeaa458bd0cc
e33491ab4e59
af693c12bcea
e8a7e01cdd11
06a58b9143a9
5fd4b8fccb71
6b5024a4ae91
0f7db9d3112a
6595c5409775
7a97bc73c6de
6db5fc269d7a
5bf64f3c0864
4d09da1f674e
```

```
[root@gpu01 ~]# docker rmi 8791df39c63e 6ec7a9230a07 6645f51ac6a2 1b3e5202a4a0
```

```
(second batch of lerobot untagged images removed)
```

```
[root@gpu01 ~]# docker images
```

aidc-artgarage:5000/multimatter-lerobot	<none>	27dede7dfab2
aidc-artgarage:5000/cleargrasp-train	latest	80be2058e2ab

```
isaac-lab-strider          latest  cf58f698fe91
aidc-artgarage:5000/ultralytics/ultralytics latest 75f29cdb37cc
aidc-artgarage:5000/chetan/tensorflow 25.02-tf2-py3 c73c55d3f8b9
nvcvcr.io/nvidia/isaac-sim 4.5.0 ca34eb4ab11f
```

2.8 Workload Check – gpu01 Still Running Jobs

```
[root@gpu01 ~]# top
top - 13:29:58 up 5:07, 1 user, load average: 4.15, 4.34, 4.46
Tasks: 1440 total, 13 running, 1427 sleeping, 0 stopped, 0 zombie
MiB Mem : 2063488 total, 1710519 free, 178744 used, 174224 buff/cache
  PID USER      %CPU %MEM    COMMAND
 29765 sujith   101.0   1.4   python
 29766 sujith   100.7   1.4   python
 29767 sujith   100.7   1.4   python
```

[i] PBS job 5279 (sujith/fiall) is actively consuming ~100% CPU per core on multiple Python processes — workload healthy and unaffected.

2.9 isaac-sim Re-tagging and Final Cleanup

```
[root@gpu01 ~]# history | grep pull
146  docker pull aidc-artgarage:5000/wtw-1:latest
242  docker pull aidc-artgarage:5000/tensorflow:25.02-tf2-py3
251  docker pull aidc-artgarage:5000/chetan/test_gpu:latest
374  docker pull aidc-artgarage:5000/wtw-1:latest
375  docker pull aidc-artgarage:5000/tensorflow:25.02-tf2-py3
376  docker pull aidc-artgarage:5000/pytorch:22.02-py3
```

```
[root@gpu01 ~]# docker tag aidc-artgarage:5000/isaac-sim ca34eb4ab11f
Error response from daemon: No such image: aidc-artgarage:5000/isaac-sim:latest
```

✗ Arguments in wrong order. docker tag syntax is: docker tag <source_id> <new_name:tag>

```
[root@gpu01 ~]# docker tag ca34eb4ab11f aidc-artgarage:5000/isaac-sim
(no output – tag created successfully)
```

```
[root@gpu01 ~]# docker images
REPOSITORY          TAG          IMAGE ID          CREATED          SIZE
isaac-lab-strider    latest       cf58f698fe91     4 months ago    36.6GB
aidc-artgarage:5000/isaac-sim latest       ca34eb4ab11f     12 months ago   15GB
nvcvcr.io/nvidia/isaac-sim 4.5.0       ca34eb4ab11f     12 months ago   15GB
```

```
[root@gpu01 ~]# docker pull aidc-artgarage:5000/isaac-sim:latest
Error response from daemon: Get "http://aidc-artgarage:5000/v2/": dial tcp 10.1.1.101:5000: connect: connection refused
```

✗ Docker registry on aidc-artgarage:5000 unreachable — push/pull not possible. Registry was down at the time; unrelated to IB fabric.

```
[root@gpu01 ~]# history | grep push
693  # docker push aidc-artgarage:5000/isaac-sim:latest
694  history | grep push
```

```
[root@gpu01 ~]# docker push aidc-artgarage:5000/isaac-sim:latest
The push refers to repository [aidc-artgarage:5000/isaac-sim]
Get "http://aidc-artgarage:5000/v2/": dial tcp 10.1.1.101:5000: connect: connection refused
```

```
[root@gpu01 ~]# docker images --no-trunc | grep ca34eb4ab11f
```

```
nvcr.io/nvidia/isaac-sim      4.5.0    sha256:ca34eb4ab11f3a9e3cb207068a66e0787e4273a7...
aidc-artgarage:5000/isaac-sim latest    sha256:ca34eb4ab11f3a9e3cb207068a66e0787e4273a7...
```

```
[root@gpu01 ~]# docker rmi nvcr.io/nvidia/isaac-sim:4.5.0
Untagged: nvcr.io/nvidia/isaac-sim:4.5.0
```

```
[root@gpu01 ~]# docker rmi aidc-artgarage:5000/isaac-sim:latest
Untagged: aidc-artgarage:5000/isaac-sim:latest
Deleted: sha256:ca34eb4ab11f3a9e3cb207068a66e0787e4273a7de92e5435681c2c6193d9849
```

```
[root@gpu01 ~]# docker images
REPOSITORY    TAG    IMAGE ID    CREATED    SIZE
(empty - all images deleted)
```

✓ All Docker images removed from gpu01.

2.10 Post-Cleanup Verification

```
[root@gpu01 ~]# qstat -ans
Jobs 5277 and 5279 still R at elapsed 02:37 and 02:25 - workloads unaffected
```

```
[root@gpu01 /]# du -sh var
125G    var
```

```
[root@gpu01 /]# du -sh var/lib/docker
63G     var/lib/docker
```

[i] Docker reduced from 456 GB → 63 GB (393 GB freed). Remaining 63 GB is the overlay2 layer cache from the completed container runs.

```
[root@gpu01 ~]# df -h
(Lustre @o2ib mount still active - IB fabric healthy throughout entire cleanup)
```

2.11 PBS mom_logs Investigation

```
[root@aidc-artgarage ~]# ssh gpu01
Last login: Fri Jan 23 14:32:05 2026 from 10.1.1.101
```

```
[root@gpu01 ~]# cd /var/spool/
(changed to /var/spool/)
```

```
[root@gpu01 spool]# ls
anacron  cron  lpd  mail  pbs  rhsm
```

```
[root@gpu01 spool]# du -sh *
12K    anacron
0      cron
0      lpd
0      mail
63G    pbs
0      rhsm
```

```
[root@gpu01 spool]# cd pbs/
(changed to /var/spool/pbs/)
```

```
[root@gpu01 pbs]# ls
aux  checkpoint  mom_logs  mom_priv  pbs_environment  pbs_version  spool  undelivered
```

```
[root@gpu01 pbs]# du -sh *
16K    aux
0      checkpoint
63G    mom_logs
2.7M   mom_priv
4.0K   pbs_environment
4.0K   pbs_version
17M    spool
0      undelivered
```

```
[root@gpu01 pbs]# cd ../../
(back to /var/)
```

```
[root@gpu01 var]# cd lib/
(changed to /var/lib/)
```

```
[root@gpu01 lib]# du -sh docker
63G    docker
```

```
[root@gpu01 lib]# cd docker
(changed to /var/lib/docker/)
```

```
[root@gpu01 docker]# ls
buildkit  containers  engine-id  image  network  overlay2  plugins  runtimes  swarm  tmp  volumes
```

```
[root@gpu01 docker]# du -sh *
4.3M   buildkit
0      containers
4.0K   engine-id
35M    image
50M    network
63G    overlay2
0      plugins
0      runtimes
0      swarm
0      tmp
24K    volumes
```

```
[root@gpu01 docker]# cd overlay2/
```

(changed to /var/lib/docker/overlay2/)

```
[root@gpu01 overlay2]# ls
```

(long listing of overlay2 layer directory hashes)

```
[root@gpu01 overlay2]# pwd
```

/var/lib/docker/overlay2

```
[root@gpu01 overlay2]# cd /var/spool/pbs/mom_logs/
```

(changed to mom_logs directory)

```
[root@gpu01 mom_logs]# ls
```

```
20250728 20250729 20250730 20250731 20250801 20250802 20250803 20250804
20250805 20250819 20250820 20250821 20250822 20250823 20250824 20250825
... (daily log files from 2025-07-28 through 2026-01-23 - ~6 months = 63 GB)
```

✗ 63 GB of PBS mom_logs going back to July 2025 included in cm image capture by default. Must be excluded.

3. Terminal Session – File 2: hpcm-vm Disk Monitoring

Source: MobaXterm_root_hpcm-vm_20260123_152500.rtf | Concurrent with gpu01 Docker cleanup.

3.1 First hpcm-vm Session – Disk Rising

```
[root@aidc-artgarage ~]# ssh hpcm-vm
```

Activate the web console with: systemctl enable --now cockpit.socket
Last login: Fri Jan 23 13:33:14 2026 from 10.1.1.101

```
[root@hpcm-vm ~]# ls
```

cluster-configfile Desktop Documents Downloads ifcfg-enp3s0-1 Music Pictures

```
[root@hpcm-vm ~]# df -h
```

Filesystem	Size	Used	Avail	Use%	Mounted on
devtmpfs	30G	0	30G	0%	/dev
tmpfs	30G	2.4M	30G	1%	/dev/shm
tmpfs	30G	66M	30G	1%	/run
tmpfs	30G	0	30G	0%	/sys/fs/cgroup
/dev/vda31	117G	80G	38G	69%	/
/dev/vda11	966M	248M	653M	28%	/boot
tmpfs	5.9G	0	5.9G	0%	/run/user/0
tmpfs	5.9G	20K	5.9G	1%	/run/user/42

i Disk at 69% (80 GB used of 117 GB) — image capture has started and is consuming space via rsync.

```
[root@hpcm-vm ~]# df -h
```

(second poll — /dev/vda31 now at 70%, Used: 82G)

```
[root@hpcm-vm ~]# watch df -h
```

(launched interactive watch loop — refreshes every 2 seconds)

3.2 Second hpcm-vm Session – Continued Monitoring

```
[root@aidc-artgarage ~]# ssh hpcm-vm
```

Activate the web console with: systemctl enable --now cockpit.socket
Last login: Fri Jan 23 13:48:39 2026 from 10.1.1.101

```
[root@hpcm-vm ~]# df -h
```

/dev/vda31 117G 69G 49G 59% / (disk dropped — old image deleted)

```
[root@hpcm-vm ~]# df -h
```

/dev/vda31 117G 77G 40G 66% / (new capture started — rising again)

```
[root@hpcm-vm ~]# df -h
```

/dev/vda31 117G 86G 32G 73% /

```
[root@hpcm-vm ~]# df -h
```

/dev/vda31 117G 99G 19G 85% / (accelerating — Docker overlay2 syncing)

```
[root@hpcm-vm ~]# df -h
```

/dev/vda31 117G 104G 14G 89% /

```
[root@hpcm-vm ~]# df -h
```

/dev/vda31 117G 105G 13G 90% /

```
[root@hpcm-vm ~]# df -h
/dev/vda31 117G 106G 12G 91% /
```

```
[root@hpcm-vm ~]# watch df -h
(another watch loop launched – monitoring rise to 100%)
```

3.3 Third hpcm-vm Session – Post-Cleanup Recovery

```
[root@hpcm-vm ~]# df -h
/dev/vda31 117G 61G 57G 52% / (Docker cleanup on gpu01 succeeded – disk recovered)
```

```
[root@hpcm-vm ~]# watch df -h
(watch launched again to monitor second capture attempt)
```

```
[root@hpcm-vm ~]# watch df -h
(watch launched third time – monitoring successful final capture)
```

✓ After Docker cleanup on *gpu01*, *hpcm-vm* disk recovered to 52%. Subsequent captures with *--exclude* completed without hitting 100%.

4. Terminal Session – File 3: HPCM Image Management on hpcm-vm

Source: MobaXterm_root_hpcm-vm_20260123_152537.rtf | Terminal session starts after a large rsync filesystem listing.

4.1 Navigate to Images Directory and Review History

```
[root@hpcm-vm images]# ls
rhel8.10      rhel8.10-gpu01-27072025
```

```
[root@hpcm-vm images]# pwd
/opt/clmgr/image/images
```

```
[root@hpcm-vm images]# history | grep delete
227  nmcli connection delete ff:feel:50c/64
357  cm image delete -i rhel8.10-gpu01-25072025
359  cm image delete -i rhel8.10-gpu01
367  cm image delete -i rhel8.10-gpu01-25072025
393  cm image delete -i rhel8.10-gpu01-27072025
406  cm image delete -i rhel8.10-gpu01-27072025
464  cm image delete -i rhel8.10-gpu01-27072025
521  history | grep delete
```

```
[root@hpcm-vm images]# history | grep capture
354  cm image capture --node gpu01 -i rhel8.10-gpu01-25072025
366  cm image capture --node gpu01 -i rhel8.10-gpu01-25072025
403  cm image capture --node gpu01 -i rhel8.10-gpu01-27072025
413  cm image capture --node gpu01 -i rhel8.10-gpu01-27072025
422  #cm image capture --node gpu01 -i rhel8.10-gpu01-29072025
432  cm image capture --node gpu01 -i rhel8.10-gpu01-27072025
526  history | grep capture
```

4.2 Pre-Capture Disk Check and Delete Old Image

```
[root@hpcm-vm images]# df -h
Filesystem      Size  Used Avail Use% Mounted on
/dev/vda31      117G   56G   62G   48% /
/dev/vda11      966M  248M  653M   28% /boot
```

[i] Disk at 48% (56 GB used) before capture attempt. Sufficient space for a clean capture — but Docker overlay2 will exceed this.

```
[root@hpcm-vm images]# cm image delete -i rhel8.10-gpu01-27072025
Removing rhel8.10-gpu01-27072025 ...
Removing any possible version control data for the image...
Configuration manager submitting node configuration.
Populating Dataset...
Populating Dataset complete: 0.262s
1 of 1 nodes completed in 2.8 seconds, averaging 0.0s per node
Node configuration complete.
recreate or delete specified, removing miniroot content for this image if it exists
Remove possible content related to NFS root support
Note: /opt/clmgr/shared_storage/image/images_dist/rhel8.10-gpu01-27072025/ does not exist
Note: /opt/clmgr/shared_storage/image/images_ro_nfs/rhel8.10-gpu01-27072025/ does not exist
Admin: Running tool to update LIO iscsi setup.
configure-iscsi: iscsi configuration - logging to /opt/clmgr/log/configure-iscsi
```

```
[root@hpcm-vm images]# cm image show rhel8.10
```


(only rhel8.10 base image listed – rhel8.10-gpu01-27072025 successfully removed)

4.3 First Image Capture Attempt – Disk Full

```
[root@hpcm-vm images]# cm image capture --node gpu01 -i rhel8.10-gpu01-23012026
Attempting to start cm_prepareclient on the remote node...
Running command: ssh gpu01 /opt/clmgr/bin/cm_prepareclient --server admin
cm_prepareclient: Using these IP(s) for hosts allow: 10.1.1.110
cm_prepareclient: Temporary rsync config file: /tmp/tmp.IZUlpYoHJ2
cm_prepareclient: Temporary rsync log file: /tmp/tmp.LiSUfZfH8x
cm_prepareclient: Giving rsync a moment to start: ...
cm_prepareclient: You may now use cm image capture on the admin node to capture this node's image.
-----
cm_prepareclient successfully started on gpu01
-----
Running rsync...
rsync: write failed on ".../var/lib/docker/overlay2/...": No space left on device (28)
rsync error: error in file IO (code 11) at receiver.c(374)
Error: rsync failed to retrieve image rhel8.10-gpu01-23012026 from gpu01
```

✗ CRITICAL FAILURE: rsync error code 11 — No space left on device. hpcm-vm /dev/vda31 hit 100% because Docker overlay2 (~456 GB) was included in the rsync operation.

```
[root@hpcm-vm images]# df -h
/dev/vda31 117G 117G 5.2M 100% / <— DISK FULL
```

4.4 CM Database Warning – Orphaned Image Cleanup (repeated cycle x4)

```
[root@hpcm-vm images]# cm image show rhel8.10
Warning: The following image(s) were found on the filesystem in
/opt/clmgr/image/images but not in the CM DB
  rhel8.10-gpu01-23012026
One or more images were not configured in all proper locations.
If an image exists in /opt/clmgr/image/images but is missing from the
database, use 'cm image create --use-existing' to import it.
Otherwise, use 'cm image delete' to delete the image entirely.
```

[i] Failed capture left an orphaned directory on the filesystem but never registered in the CM database. Must use cm image delete -i to purge it before retrying.

```
[root@hpcm-vm images]# cm image delete -i rhel8.10-gpu01-23012026
Warning: image is on the file system but not in the database, Attempting to delete
Removing any possible version control data for the image...
/opt/clmgr/image/miniroot/pre_squeeze/rhel8.10-gpu01-23012026 does not exist
/opt/clmgr/image/miniroot/squeezed/rhel8.10-gpu01-23012026 does not exist
Note: /opt/clmgr/shared_storage/image/images_dist/rhel8.10-gpu01-23012026/ does not exist
Admin: Running tool to update LIO iscsi setup.
configure-iscsi: iscsi configuration - logging to /opt/clmgr/log/configure-iscsi
```

```
[root@hpcm-vm images]# df -h
/dev/vda31 117G 56G 62G 48% / (disk recovered after orphan removed)
```

[i] This cm image capture → disk full → cm image delete cycle repeated 4 times before the --exclude solution was found.

4.5 Navigate to Images Dir from /opt and Retry Attempts

```
[root@hpcm-vm images]# cd
(returned to home directory)
```

```
[root@hpcm-vm ~]# ls
cluster-configfile  Desktop  Documents  Downloads  ifcfg-enp3s0-1  Music  Pictures
```

```
[root@hpcm-vm ~]# cd /opt/
(changed to /opt/)
```

```
[root@hpcm-vm opt]# ls
c3  clmgr  clustertest  cmu  fluent-bit  oscar  sgi
```

```
[root@hpcm-vm opt]# cm image show rhel8.10
(base image only — database clean after previous delete)
```

```
[root@hpcm-vm opt]# cm image capture --node gpu01 -i rhel8.10-gpu01-23012026 --e...
(truncated — another capture attempt; also failed with disk full)
```

```
[root@hpcm-vm opt]# cm image show rhel8.10 rhel8.10-gpu01-23012026
(image registered — brief window before rsync failure fills disk again)
```

```
[root@hpcm-vm opt]# ls
c3  clmgr  clustertest  cmu  fluent-bit  oscar  sgi
```

```
[root@hpcm-vm opt]# cd clmgr/image/images
(changed to /opt/clmgr/image/images/)
```

```
[root@hpcm-vm images]# ls
rhel8.10  rhel8.10-gpu01-23012026
```

```
[root@hpcm-vm images]# cd rhel8.10-gpu01-23012026/
(entered captured image root filesystem to verify contents)
```

```
[root@hpcm-vm rhel8.10-gpu01-23012026]# ls
bin  boot  dev  etc  home  lib  lib64  media  mnt  opt  proc  root  run  sbin  srv  sys  tmp  usr
var
```

```
[root@hpcm-vm rhel8.10-gpu01-23012026]# cd var/spool/
(navigated into captured image's /var/spool/)
```

```
[root@hpcm-vm spool]# ls
anacron  cron  lpd  mail  rhsm
```

[i] Verification: pbs directory is absent from the captured image's /var/spool/ — confirming mom_logs were excluded.

4.6 Successful Capture with --exclude Flags (main solution)

```
[root@hpcm-vm ~]# cm image show rhel8.10
(base only – clean state before final attempt)
```

```
[root@hpcm-vm ~]# cm image capture --node gpu01 -i rhel8.10-gpu01-23012026 --exclude
/var/spool/pbs/mom_logs/* --exclude /var/lib/docker/overlay2/*
Attempting to start cm_prepareclient on the remote node...
Running command: ssh gpu01 /opt/clmgr/bin/cm_prepareclient --server admin
cm_prepareclient: Using these IP(s) for hosts allow: 10.1.1.110
cm_prepareclient: Temporary rsync config file: /tmp/tmp.otTRks0kNR
cm_prepareclient: Temporary rsync log file: /tmp/tmp.nfiMlsHk9X
cm_prepareclient: Giving rsync a moment to start: ...
cm_prepareclient: You may now use cm image capture on the admin node to capture this node's image.
cm_prepareclient: Note: this script backgrounds, see system logging for debugging.
-----
cm_prepareclient successfully started on gpu01
-----
[rsync runs successfully – no disk full error]
```

✓ *SUCCESS: cm image capture completed. Excluding /var/spool/pbs/mom_logs/* and /var/lib/docker/overlay2/* resolved the disk exhaustion.*

```
[root@hpcm-vm ~]# cm image show rhel8.10 rhel8.10-gpu01-23012026
(both images present in CM database – capture confirmed successful)
```

4.7 Verify Captured Image Contents

```
[root@hpcm-vm pbs]# ls
aux mom_logs pbs_environment spool checkpoint mom_priv pbs_version undelivered
```

[i] The successful capture does include the /var/spool/pbs/ directory structure (aux, checkpoint, mom_priv etc.) — only mom_logs/ was excluded as intended.*

```
[root@hpcm-vm docker]# ls
buildkit engine-id network plugins swarm volumes containers image overlay2 runtimes tmp
```

[i] The captured image retains /var/lib/docker/ directory structure and metadata — only the overlay2/ layers were excluded.*

```
[root@hpcm-vm docker]# cm image show rhel8.10 rhel8.10-gpu01-23012026
rhel8.10
rhel8.10-gpu01-23012026
```

4.8 Final Image Deletion (cleanup / re-capture preparation)

```
[root@hpcm-vm ~]# cm image delete -i rhel8.10-gpu01-23012026
```

```
Removing rhel8.10-gpu01-23012026 ...  
Removing any possible version control data for the image...  
Configuration manager submitting node configuration.  
Populating Dataset...  
Populating Dataset complete: 0.281s  
0 of 1 nodes completed in 2.8 seconds, averaging 0.0s per node  
1 of 1 nodes completed in 5.3 seconds, averaging 0.0s per node  
Node configuration complete.
```

```
[root@hpcm-vm ~]# cm image show rhel8.10
```

```
(only base rhel8.10 listed - gpu01-23012026 cleanly removed)
```

5. All Errors and Resolutions

Error (verbatim from terminal)	Root Cause	Resolution	Result
rsync: write failed: No space left on device (28) rsync error: code 11 at receiver.c(374)	Docker overlay2 (~456 GB) and PBS mom_logs (~63 GB) included in rsync to hpcm-vm 117 GB disk	Clean Docker on gpu01 first; use --exclude /var/spool/pbs/mom_logs/* --exclude /var/lib/docker/overlay2/*	Resolved
conflict: unable to delete c73c55d3f8b9 (must be forced)	Same image ID shared by two named tags (tensorflow + chetan/tensorflow)	docker images --no-trunc grep <id> to find all tags; docker rmi each tag name separately	Resolved
conflict: unable to remove — container is using image	14 stopped containers still referenced images; Docker blocks removal	docker ps -a to list; docker stop <ids>; docker rm \$(docker ps -aq); retry docker rmi	Resolved
docker tag aidc-artgarage:5000/isaac-sim ca34eb4ab11f: No such image	Arguments reversed — destination given before source ID	Correct: docker tag <source_id_or_name> <destination:tag>	Resolved
"docker ps" accepts no arguments (docker ps a / docker ps aa)	Missing hyphen — 'a' treated as positional argument not a flag	Correct flag: docker ps -a	Resolved
qstat: illegally formed job identifier: 5277.aidc-artg*	Wildcard (*) not valid in qstat job identifier	Use full numeric ID: qstat -xf 5277	Resolved
du: cannot access '-': No such file or directory (du - sh)	Space between '-' and 'sh' splits the flag into two arguments	Correct: du -sh <path>	Resolved
cm image show: Warning — image found on filesystem but not in CM DB	Failed capture created /opt/clmgr/image/images/rhel8.10-gpu01-23012026 dir but never completed DB registration	cm image delete -i rhel8.10-gpu01-23012026 to purge orphaned filesystem entry; disk recovers to 48%	Resolved (x4 times)
docker push/pull: dial tcp 10.1.1.101:5000: connect: connection refused	Docker registry on aidc-artgarage:5000 was down/unreachable	Registry was unavailable at that time; unrelated to IB fabric	Not resolved (registry issue)

6. InfiniBand Infrastructure Observations

The IB switch and fabric caused no issues in this session. Every `df -h` on `gpu01` confirmed Lustre availability throughout all phases.

```
[root@gpu01 ~]# df -h (IB/Lustre line - present in every df output)
10.128.1.202@o2ib,10.128.1.203@o2ib:10.128.1.204@o2ib,10.128.1.205@o2ib:/lustre1 (mounted)
```

[i] @o2ib confirms RDMA over InfiniBand. All four Lustre storage servers reachable. `mlx5_core` driver confirmed active in `gpu01` miniroot `initrd`.

IB Component	Evidence from Session
mlx5_core + mlx4	Miniroot <code>initrd</code> config lists: <code>mlx5_core</code> , <code>mlxfw</code> , <code>mlx4_en</code> , <code>mlx4_core</code>
UCX/RDMA stack	<code>libucx</code> , <code>libuct</code> , <code>libucp</code> , <code>libibverbs</code> installed in <code>/opt</code> on <code>gpu01</code>
Lustre @o2ib	Present in every <code>df -h</code> output throughout the entire 3-hour session
lldpad daemon	Visible in <code>top</code> output on <code>aidc-artgarage</code> (Link Layer Discovery Protocol)
PBS RDMA comms	Uninterrupted PBS job execution over IB while Docker cleanup ran concurrently